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# DDM-YOLOv8s for Small Object Detection in Remote Sensing Images

line 1: 1st Yangli Xi
line 2: *dept. School of Cyberspace Security,Zhengzhou University*
line 4: Zhengzhou, Henan, China
line 5: 1141217642@qq.com

line 1: 2nd Dan Qu
line 2: *dept. School of Information Systems Engineering,Information Engineering University*
line 4: Zhengzhou, Henan, China
line 5: qudanqudan@163.com*

line 1: 3rd Liming Du
line 2: *dept. School of Cyberspace Security,Zhengzhou University*
line 4: Zhengzhou, Henan, China
line 5: d192102663@163.com

***Abstract*—In order to solve the problem of low accuracy of remote sensing target detection caused by insufficient feature representation, complex background and diversified scales in remote sensing image target detection task, a remote sensing image target detection algorithm based on DDM-YOLOv8s (DySample DASI-MODSConvYOLOv8s) model was proposed. First, a super lightweight and efficient dynamic upper sampler DySample is used to improve the image resolution of the algorithm without adding additional burden. Secondly, the new C2f_DASI module is constructed by using the dimensional awareness selective integration module DASI and C2f, and the adaptive channel selection and fusion are realized in the feature fusion module, which effectively solves the complex background problem in remote sensing images; Finally, the head part is redesigned by Maintaining the Original information - Depth Separable Convolution MODSConv, which maintains as much feature layer information as possible before prediction, so that the extracted features contain more correlation information, greatly improving the detection accuracy and speed. The experimental results on RSOD datasets show that the test mAP of the improved YOLOv8s algorithm is 94.5%, which proves that this method can effectively improve the detection accuracy of small targets in remote sensing images.***

***Keywords*—remote sensing image, YOLOv8s, Dynamic upper Sampler DySample, Dimension awareness selective integration module DASI, The original information preserving depth separable convolution MODSConv is adopted***

## I. INTRODUCTION

Remote sensing image target detection is an important research direction in the field of geographic information technology. It plays an important role in environmental detection, geological exploration, vegetation detection, marine ecological monitoring, surface disaster monitoring and other fields.With the development of computer vision technology, Convolution Neural Network (CNN) has been applied to target recognition in remote sensing images The key difference of target detection techniques is how to extract features from images In the field of remote sensing, target detection usually depends on the existing main algorithms, which can be divided into two categories: two-stage algorithm and single-stage algorithm Two-stage target detection firstly generates a group of candidate regions from the input image, and then extracts and classifies these regions. Common two-stage target detection algorithms such as R-CNN series algorithms[1] 。 Although the two-stage detection algorithm is excellent in accuracy, it is difficult to meet the urgent needs of real-time detection of remote sensing images because of its complex training and slow detection speed. Single stage algorithms include YOLO (You Only Look Once)[2]Series and SSD (Single Shot MultiBox Detector)[3].The core idea of this method is to use direct regression technology to determine the location and category information of the target, which greatly increases the computational demand and significantly accelerates the detection speed However, the improvement of efficiency is at the expense of certain detection accuracy In order to enhance the detection ability of YOLO series algorithm in the case of complex background, low-scale, diverse and dense occlusion, it is necessary to further optimize the algorithm.The main methods are: improving the feature iteration of the backbone network to enhance feature extraction[4]; Integrate attention mechanism and highlight target feature weight[5]; Through DenseConnection, enhance information transmission and sharing between levels, and expand the reuse range of features[6]; Optimize the anchor frame mechanism[7], loss function[8]; Using adaptive pruning compression algorithm to improve reasoning speed[9]; Improve NMS (Non Maximum Suppression) algorithm, optimize target area screening[10], etc.

In the research of remote sensing image target detection, because of its small pixel coverage area, its subtle features are easily ignored or lost in complex background after multiple

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convolution and pooling operations by deep network. By fusing attention mechanism, it is still difficult to accurately lock the position and feature information of the region of interest in highly complex scenes, which limits its performance in small target detection in complex environments. However, traditional up sampling methods, such as interpolation algorithm, can only estimate the value of new pixels based on the pixel values around the pixels, which leads to the loss of small target detail features in the up sampling process, and it is difficult to restore the original detail features, thus affecting the accuracy of subsequent detection algorithms. In YOLO algorithm, Convolution is one of the core components, which is responsible for extracting image features. However, there are many problems such as large amount of computation, many parameters and inefficient extraction of key features in the image. The convolution operation itself is computationally intensive, especially when processing high-resolution images, which will lead to slower processing speed and affect system performance. Depth Separable Convolution Technique is used to decompose the convolution operation into two stages: depth convolution and point convolution, which greatly increases the number of parameters and computational complexity and improves the processing efficiency However, this decomposition also brings about the complexity of the model, which makes the model face greater challenges in training and optimization.

In conclusion, aiming at the above problems, this paper proposes a remote sensing image target detection algorithm based on DDM-YOLOv8s (DySample DASI-MODSConvYOLOv8s) model. The main work is as follows: (1) Using a super lightweight and efficient dynamic upper sampler DySample, DySample requires less computing resources, so that the algorithm can improve image resolution without adding additional burden. (2) The new C2f_DASI module is constructed by using the dimensional awareness selective integration module DASI and C2f. The adaptive channel selection and fusion are realized in the feature fusion module, which effectively solves the complex background problem in remote sensing images. (3) The header structure is innovative reformed by implementing a strategy called deep separable convolution MODSConv This strategy is dedicated to preserving the original information of the feature layer to the maximum extent before the prediction stage, so that the extracted features are more relevant and rich Therefore, this improvement greatly enhances the accuracy of detection and improves the execution speed of detection at the same time.

## II. Target detection algorithm of remote sensing image based on DDM-YOLOv8s

### *A. Improved DDM-YOLOv8s algorithm*

YOLOv8 is a target detection algorithm, which is improved and optimized on the basis of YOLO series of algorithms. It is an algorithm released by Ultralytics on January 10, 2023. Compared with YOLOv5[11]and YOLOv7[12], YOLOv8 has improved detection accuracy and speed, mainly including three components: backbone network (Backbone), neck structure (Neck) and head prediction layer (Head). YOLOv8 algorithm improves the speed and accuracy of target detection, but there are still some problems for small target detection in remote sensing images with complex background. Based on this, this paper designs a DDM-YOLOv8s (DySample DASI-MODSConvYOLOv8s) algorithm, and improves YOLOv8s algorithm to improve the performance of small target detection.

In the remote sensing image target detection task, a remote sensing image target detection algorithm based on DDM-YOLOv8s (DySample DASI-MODSConvYOLOv8s) model is proposed to solve the problem that the small target detection accuracy is not high due to insufficient feature representation, complex background and diversified scales in the remote sensing image target detection task. Its network structure is shown in Figure 1. First of all, a super lightweight and efficient dynamic upper sampler DySample is used. DySample requires less computing resources, so that the algorithm can improve image resolution without adding additional burden. Secondly, use the dimensional awareness selective integration module DASI and C2f to build a new C2f_DASI module, replace all C2f modules in the Neck network, realize adaptive channel selection and fusion in the feature fusion module, effectively solve the complex background problem in remote sensing images. Finally, the header structure is reconceived and the depth separable convolution MODSConvv is introduced to preserve more feature layer information before prediction and solve the information interaction problem between channels with lightweight These improvements make the extracted features more relevant, which significantly improves the detection accuracy and speed MMD-YOLOv8s algorithm effectively improves the detection accuracy of small targets in complex background through these optimization measures.

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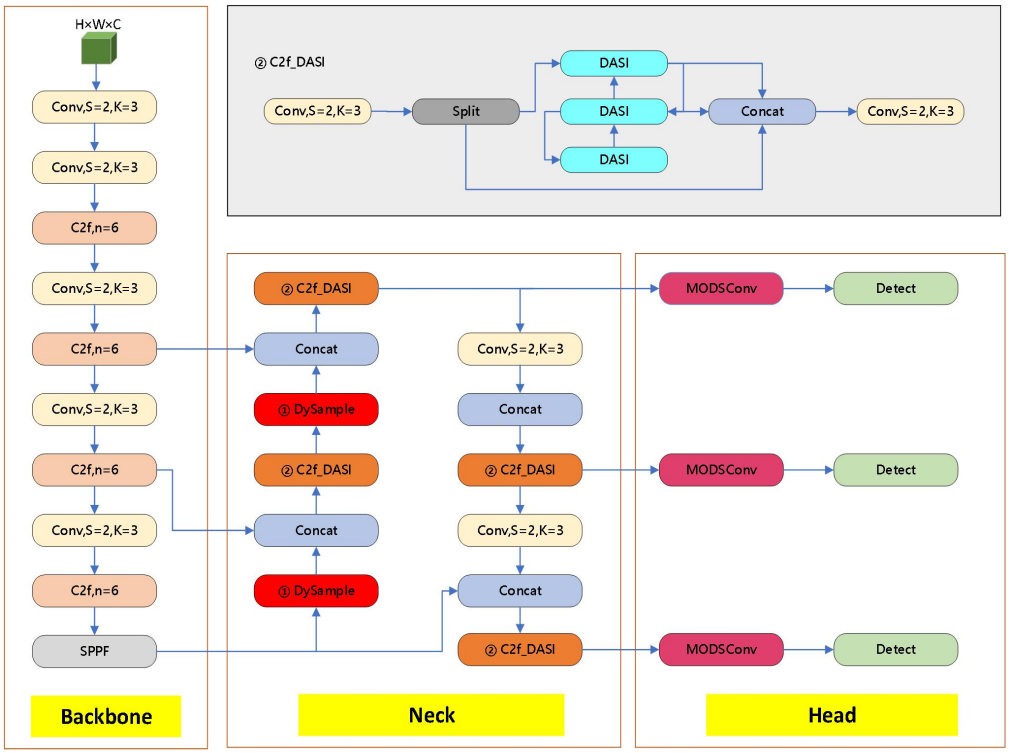



Fig. 1. Network Structure of DDM-YOLOv8s

## B. *DySample, DASI and MODSConv*

DySample is an ultra-lightweight and efficient dynamic up-sampler Kernel-based dynamic up-samplers such as CARAFEFADE and SAPA have significant workload because they involve a large number of dynamic convolution operations and generate additional subnets of dynamic kernels In addition, the dependence of FADE and SAPA on high-resolution features limits their application to some extent To overcome these challenges, DySample designed a more efficient up-sampling from the point sampling perspective, which can be easily implemented with the built-in functions in PyTorch without customization of CUDA packages DySample shows less resource requirements in terms of parameter number, Floating-point Operations (FLOPs), GPU memory footprint and latency while system performance

DASI, A dimension awareness selective integration module can realize adaptive channel selection and fusion, and its detailed structure is shown in Figure 2. In multiple down sampling stages of infrared small target detection, high-dimensional features may lose information about small targets, while low-dimensional features may not provide sufficient context. In order to solve this problem, a new channel division selection mechanism (as shown in Figure 2) is proposed, which enables DASI to adaptively select appropriate fusion features according to the size and features of objects.

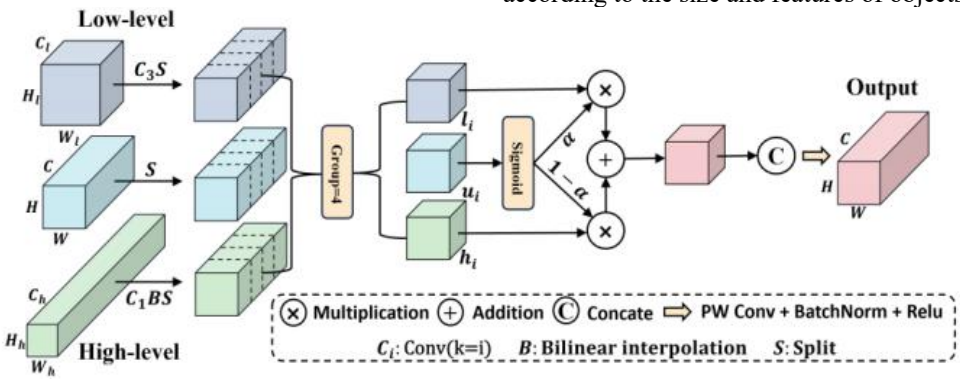



Fig. 2. Detail structure of the dimension aware selective integration module

MODSConv, i.e. Depth separable convolution preserving original information, can overcome the limitation of information interaction between channels in the original feature layer, which is similar to the classical depth separable convolution DWConv The features extracted by MODSConv not only contain richer correlation information but also retain as much feature layer information as possible before prediction and also have lightweight features This improvement

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significantly improves the detection accuracy and speed of small targets in complex background

## III. Experimental results and analysis

### A. Experimental environment and parameter settings

The experimental environment of this article is configured as follows: NVIDIA GeForce RTX4090, 24GB video memory, and Ubuntu 20.04 operating system. The programming language used is python3.8, CUDA is 11.3, YOLOv8s dependency library ultralytics version is 8.0.157, the total number of model training iterations is 200, the initial learning rate is set to 0.01, and the batch size is 16. The linear learning rate attenuation strategy is adopted, and the optimizer is a random gradient decline SGD[13].

### B. The experimental datasets

RSOD datasets is adopted. RSOD is an open target detection data set for target detection in remote sensing images. The data set includes aircraft, fuel tanks, stadiums and overpasses. Each category includes 4993 aircraft in 446 images, 191 playgrounds in 189 images, 180 overpasses in 176 images, and 1586 fuel tanks in 165 images.

### C. Public evaluation indicators

The experimental results in this paper use the commonly used evaluation indicators of target detection: accuracy (P), Recall (R), Parameters, and Floating Point Operations Per Second (FLOPs) and Mean Average Precision (mAP). MAP50 and mAP50-95 are used to represent the average precision under different IoU thresholds. The calculation formula (1) - (4) is as follows:

$$Precision = \frac{TP}{TP + FP}, \quad (1)$$

$$\mathrm{Re}call = \frac{TP}{TP + FN}, \quad (2)$$

$$AP = \int_0^1 P(R)dR, \quad (3)$$

$$mAP = \frac{\sum_{i=1}^{n} AP_i}{n}, \quad (4)$$

In the above formula, TP (True Positive): the true example means that the target in the picture is recognized as the correct target. FP (False Positive): false positive example, which means that the location of the target can be identified, but the target's category identification is wrong. FN (False Negative): false counterexample. The correct target is not identified and is considered as other targets, that is, missed detection. N represents the number of categories, and AP represents the area of P-R curve. The larger the area, the better the classifier. MAP refers to the average value of multiple categories of APs.

### D. Ablation experiments

In order to verify the performance of each module, a series of ablation experiments were designed under the RSOD dataset with YOLOv8s original model as the benchmark method. Recall, mAP50, and mAP50-95 were used as quantitative evaluation indicators. The experimental results are shown in Table 1. It can be seen from the data in the table that when using the dynamic upper sampler DySample alone, the experimental results show that Recall is increased by 1.1%, mAP50 and mAP50-95 are increased by 2% and 2.4% respectively compared with baseline; Compared with the complete model without dynamic upper sampler DySample, the mAP50 is increased by 1.6%. It shows that DySample can effectively solve the problem of low detection accuracy of small targets in complex background. When the dimensional awareness selective integration module DASI is used alone to improve the neck feature fusion end, the experimental results show that Recall is increased by 4.4%, mAP50 and mAP50-95 are increased by 3.8% and 3.5% respectively compared with baseline;. When used alone, the original information preserving depth separable convolution MODSConv is adopted, and its mAP50 and mAP50-95 are increased by 2% and 2.4% respectively compared with baseline.

In conclusion, when using the complete model, Recall, mAP50, and mAP50-95 all reach the maximum value, and Parameters and FLOPs all reach the minimum value, which indicates that DDM-YOLOv8s algorithm effectively improves the detection accuracy of small targets in remote sensing images and realizes the lightweight of the model. The proposed DDM-YOLOv8s network detection effect is good.

TABLE I. The name of the table shall be the corresponding explanatory text of the table

| YOlOv8s | DySample | C2f-DASI | MODSConv | Recall | mAP50 | mAP50-95 | Parameters/M | Computational/GFLOPs |
|---|---|---|---|---|---|---|---|---|
| √ | | | | 85.6% | 89.6% | 61.6% | 11.14 | 28.7 |
| √ | | | √ | 87.7% | 91.6% | 64.0% | 9.46 | 22.2 |
| √ | √ | | | 90.4% | 92.4% | 64.2% | 11.16 | 28.7 |
| √ | | √ | | 90.0% | 93.4% | 65.1% | 10.18 | 25.5 |
| √ | √ | | √ | 89.2% | 91.7% | 63.8% | 9.49 | 22.2 |
| √ | | √ | √ | 91.0% | 92.9% | 65.4% | 11.39 | 28.9 |
| √ | √ | √ | | 88.0% | 90.7% | 64.0% | 10.21 | 25.6 |
| √ | √ | √ | √ | **91.5%** | **94.5%** | **64.9%** | **8.53** | **19.1** |

### E. Contrasting experiments

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In order to verify the advancement of the DASI attention mechanism used, this paper conducts experiments on the RSOD dataset based on YOLOv8 by integrating a variety of different attention mechanisms. The experimental results are shown in Table 2. From the table, we can see that the DASI attention module algorithm used in this paper has the highest improvement in the detection accuracy of small targets in remote sensing images and the smallest number of model parameters on the same baseline, which has the most superior performance.

TABLE II. COMPARISON OF EXPERIMENTAL RESULTS USING DIFFERENT ATTENTIONAL MECHANISMS

| Attention mechanisms | Recall | mAP50 | mAP50-95 | Parameters/M | GFLOPs/B |
|---|---|---|---|---|---|
| YOLOv8s(baseline) | 85.6% | 89.6% | 61.6% | 11.14 | 28.7 |
| YOLOv8s+CAMixing | 87.5% | 91.7% | 65.7% | 14.51 | 31.4 |
| YOLOv8s+CAFM | 81.8% | 85.9% | 62.4% | 12.37 | 29.6 |
| YOLOv8s+PPA | 86.5% | 90.6% | 65.4% | 19.80 | 35.0 |
| YOLOv8s+CAA | 88.9% | 92.6% | 62.0% | 10.60 | 26.6 |
| YOLOv8s+MSFE | 91.1% | 92.7% | 65.7% | 12.89 | 30.1 |
| YOLOv8s+OSRAA | 80.9% | 91.6% | 63.3% | 11.93 | 29.3 |
| YOLOv8s+DASI | **90.0%** | **93.4%** | **65.1%** | **10.18** | **25.5** |

### *F. Visualization and analysis of experimental results*

The detection results on the RSOD datasets show that the improved DDM-YOLOv8s algorithm in this paper has significantly improved the performance of YOLOv8 algorithm, in which mAP50 has increased by 4.9%. In particular, the mAP50 of the three categories of aircraft, oilbank, and playground are all above 95%. The visualization result P-R curve is shown in Figure 3 below, which intuitively shows that the algorithm in this paper has significant advantages in solving the problem of small target detection in large-scale remote sensing images.

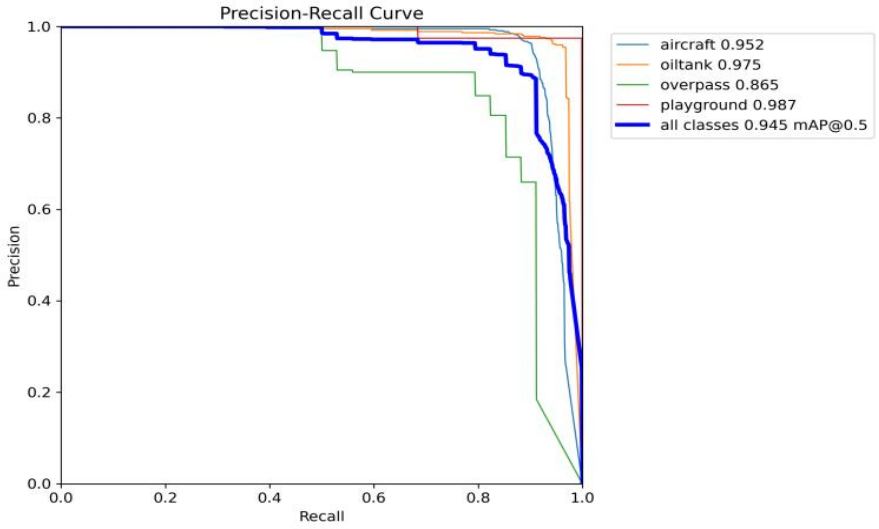
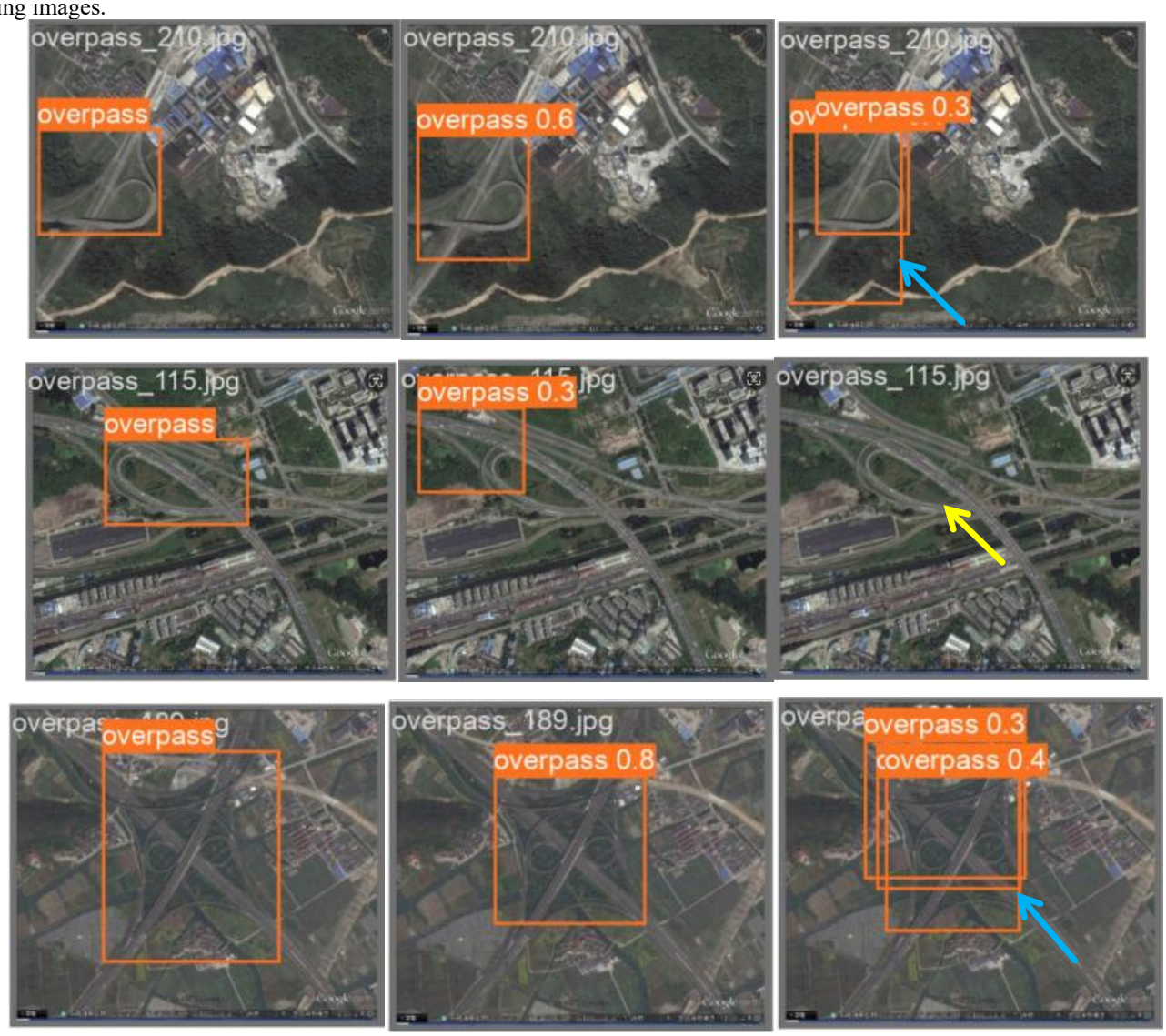



Fig. 3. P-R Curve

In order to display the detection performance of the algorithm in this paper more intuitively and vividly, we select some images of overlay in RSOD datasets with a large number of small targets and no obvious features for visualization experiments, as shown in Figure 4. In Figure 3, for the same group of images, the input images and labels, DDM-YOLOv8s (the algorithm model in this paper) and the original YOLOv8 detection results are visually compared. Among them, the blue arrow marks the false detection part of the original YOLOv8, and the yellow arrow marks the missing part of the original YOLOv8. It can be clearly observed from the figure that DDM-YOLOv8s can effectively correct these errors and improve the detection accuracy. This shows that the DDM-YOLOv8s algorithm proposed in this paper can effectively

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improve the detection performance of small targets in remote sensing images.




(a) Input the image and its label (b) YOLOv8s detection result (c) DDM-YOLOv8s detection result

Fig. 4. Comparison of partial target detection results of DDM-YOLOv8s and YOLOv8s in RSOD dataset

## IV. Conclusions

Aiming at the problems of insufficient feature representation, complex background and diversified scales in remote sensing image target detection task, this paper proposes a remote sensing image target detection algorithm based on DDM-YOLOv8s (DySample DASI MODSConvYOLOv8s) model. First of all, a super lightweight and efficient dynamic upper sampler DySample is used. DySample requires less computing resources, so that the algorithm can improve image resolution without adding additional burden. Secondly, the dimension awareness selective integration module DASI and C2f are used to build a new C2f_DASI module, replacing all C2f modules in the Neck network. The feature fusion module realizes adaptive channel selection and fusion, effectively solving the complex background problem in remote sensing images; Finally, we re-conceive the header structure by applying the Depth Separable Convolution (MODSConv) which preserves the original data, aiming at preserving the feature layer information to the maximum extent before the prediction stage and adopting a lightweight strategy to solve the problem of information interaction between channels in the original feature layer (for example, the classical depth separable convolution) These improvements significantly enhance the correlation of extracted features, thus greatly improving the accuracy and efficiency of detection. The results of ablation experiment and comparison experiment show that the improved modules have improved the detection accuracy to a certain extent, and the combination of the three improved modules has the best detection effect; Compared with other representative improved algorithms published by YOLOv8, it is obvious that the improved algorithm proposed in this paper has certain advantages in small target detection accuracy and

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detection speed under complex background, and the detection performance has been improved.

The DDM-YOLOv8 proposed in this paper has been enhanced to a certain extent in targeting the background complexity of target detection in remote sensing images, improving the model's ability to capture target features, but there are still some limitations. Based on the current research status, the following two main research directions are followed:

(1) The target detection algorithms realized in this paper are all based on the anchor frame, and one of the shortcomings of the anchor frame-based model is that it will take up a part of the time when performing NMS, especially for the scene where the target objects are densely distributed. The next step will be to explore the target detection based on no anchor frame.

(2) The next research direction will continue to study the relationship between image feature complexity and network model parameters and network model performance, if we get the expression of the relationship between them, which is of great significance to guide the selection of different sizes of network models for different scenes and tasks.

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## REFERENCES


[1] REDMON J, DIVVALAS S, GIRSHICK R, et al. You only look once: Unified, real-time object detection[C]//Proceedings of the IEEE conference on computer vision and pattern recognition.2016:779-788.

[2] LIU W, ANGUELOV D, ERHAN D, et al. Ssd: Single shot multibox detector[C]//Computer Vision−ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11−14, 2016, Proceedings, Part I 14. Springer International Publishing,2016:21-37.

[3] CARION N, MASSA F, SYNNAEVE G, et al. End-to-end object detection with transformers[C]//European conference on computer vision. Cham: Springer International Publishing,2020:213-229.

[4] XU D, WU Y. FE-YOLO: A feature enhancement network for remote sensing target detection[J]. Remote Sensing,2021,13(7):1311.

[5] CHEN L, SHI W, DENG D. Improved YOLOv3 based on attention mechanism for fast and accurate ship detection in optical remote sensing images[J]. Remote Sensing, 2021, 13(4): 660.

[6] XU D, WU Y. Improved YOLO-V3 with DenseNet for multi-scale remote sensing target detection[J]. Sensors,2020,20(15):4276.

[7] ZHAO Y Q, JIA J L, GONG W J, et al. Multi scale aerial image target detection algorithm based on pro-YOLOv4 [J] Application Research of Computers,2021,38(11):3466-3471.

[8] GEVORGYAN Z. SIoU Loss: more powerful learning for bounding box regression[J/OL]. Computer Science,2022,https://arxiv.org/abs/2205.12740.

[9] WANGJ J, WEI J, MEI S H, et al. Improved Yolov3 for small object detection in remote sensing image [J] Computer Engineering and Applications,2021,57(20): 133-141

[10] XU Z, XU X, WANG L, et al. Deformable ConvNet with aspect ratio constrained NMS for object detection in remote sensing imagery[J]. Remote Sensing,2017,9(12):1312.

[11] XU C, MAKIHARA Y, LI X, et al. Cross-view gait recognition using pairwise spatial transformer networks[J]. IEEE Transactions on Circuits and Systems for Video Technology,2020,31(1):260-274.

[12] LI G, SHI G, JIAO J. YOLOv5-KCB: A New Method for Individual Pig Detection Using Optimized K-Means, CA Attention Mechanism and a Bi-Directional Feature Pyramid Network[J]. Sensors,2023,23(11):5242.

[13] JIAXIN CAI, YANG LI, BAICHEN LIU, et al. Developing deep LSTMs with later temporal attention for predicting COVID-19 severity, clinical outcome, and antibody level by screening serological indicators over time. IEEE Journal of Biomedical and Health Informatics. 2024, 28(7):4204-4215.


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